Randomness and probability in CS

- Randomized algorithms
- Data structures using randomness
- Modeling real-world phenomena
- Machine Learning

Def Given two sets A, B

1. Sum Rule:

If $A \cap B = \emptyset$, then $|A \cup B| = |A| + |B|$

2. Product rule:

The number of pairs (x, y) with $x \in A$, $y \in B$ is $|A| \cdot |B|$

$$|A \times B| = |A| \cdot |B|$$

Ex: Suppose you have a closet with

Shirts = {Red Shirt, Blue Shirt, Black Shirt}

and

Pants = {Jeans, Khakis}

1. What is the total number of unique outfits (shirts and pants) that you can create?

$$|\text{Shirts} \times \text{pants}| = |\text{Shirts}| \cdot |\text{Pants}|$$

$$= 3 \times 2 = 6$$

(product rule)

2. What is the total number of Independent clothing options?

$$|\text{Shirts} \cup \text{pants}| = |\text{Shirts}| + |\text{Pants}|$$

$$= 3 + 2$$

(sum rule)

The reason that we can do this is Shirts $\cap$ pants $= \phi$

More generalized product rule:

$$|A_1 \times A_2 \times A_3 \times \ldots \times A_k| = |A_1| \cdot |A_2| \cdot |A_3| \cdots \cdot |A_k|$$

Inclusion-Exclusion Rule

$$|A \cup B| = |A| + |B| - |A \cap B|$$

If $A \cap B = \phi$, this rule becomes the sum rule.

Ex: $ODD = \{1, 3, 5, 7, 9\}$

$Primes = \{2, 3, 5, 7\}$

$$|ODD \cup Primes| = |ODD| + |Primes| - |ODD \cap Primes|$$

$$= 5 + 4 - 3 = 6$$

$ODD \cup Primes = \{1, 2, 3, 5, 7, 9\}$

Ex: How many length 32-bit binary strins are there.

Example binary string: 1101101110

32-bit binary string must 32 binary characters.

$\{0,1\}$

$\_\ \_\ \_\ \_\ \_\ \_\ \ldots \ldots \_$

$B = \{0,1\}$

$\langle \underline{0}, \underline{1}, \ldots, \ldots \rangle$

B^{32} = set of all 32-bit binary strings

$$|B^{32}| = |B|^{32} = 2^{32}$$

$$\underline{2} \times \underline{2} \times \underline{2} \ldots \ldots \times \underline{2} = 2^{32}$$

Popup quiz 10

1. MAC address is a unique identifier assigned to network devices. This address consists of 12 digits. Each of these 12 digits is a hexa-decimal digit.

What is the total # of unique MAC addresses we can have?

Ex: 0A:12:3B:AC:9F:11

H = Hexa-Decimal Digits = {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F}

$$MAC\_ADDRESSES = H \times H \times \ldots \ldots \times H$$

$$= H^{12}$$

$$|H^{12}| = |H|^{12} = 16^{12}$$

Q: We define a PIN to be a string with 4 digits. Each of these digits are decimal digits.

An invalid PIN would be a PIN with starting with 3-repeated digits or ending with 3-repeated digits.

How many invalid pins are there?

Ex: 0002 → 0-

4443

5555

S to be the invalid pins with repeated starting digits. $|S| = |D \times D| = 10 \cdot 10$

E to be the invalid pins with repeated ending digits. $|E| = |D \times D| = 10 \cdot 10$

$$|S \cup E| = |S| + |E| - |S \cap E|$$

$|S \cap E|$ → 10

$$= 10 \cdot 10 + 10 \cdot 10 - 10$$

$$= 190$$

Def Given some random process, the sample space S is the set of all possible outcomes.

A probability function $Pr: S \to \mathbb{R}$ and describes the fraction of the time that $s \in S$ occurs.

we have 2 rules

1. $\sum_{s \in S} Pr[s] = 1$

2. $\forall s \in S : Pr[s] \geq 0$

Example: Random process of flipping a fair coin.

$S = \{H, T\}$

$Pr[H] = 0.5 \geq 0$

$Pr[T] = 0.5 \geq 0$

$\sum_{s \in S} Pr[s] = 0.5 + 0.5 = 1$

Example: Drawing a card from a 52-card deck.

$$S = \{A\clubsuit, 2\clubsuit, \ldots\ldots, A\heartsuit, 2\heartsuit, \ldots, A\spadesuit, 2\spadesuit, \ldots, A\diamondsuit, \ldots\}$$

$$\forall s \in S: \Pr[s] = 1/52$$

Ex: Consider the random process of flipping 2 coins.

$$S = \{H, T\} \times \{H, T\}$$
$$= \{(H,H), (H,T), (T,H), (T,T)\}$$

$\Pr[(H,H)] = 1/4$ uniform probability

$\Pr[s] = 1/4, \forall s \in S$

Let $C = \{0, 1, 2, 3, 4, 5, 6, 7\}$

Choose from C by flipping 7 coins and counting the # heads.

$S = \{H, T\}^7$

$\langle H, H, H, H, H, H, H \rangle \quad 7$

$\langle T, H, H, H, H, H, H \rangle \quad 6$

$\langle T, T, T, T, T, T, T \rangle \quad 0$

$\Pr[C = 7] = 1/2^7$

$\Pr[C = 4] = 35/128$

$\langle H, H, H, H, T, T, T \rangle \rightarrow 4$

$\langle H, T, H, H, T, H, T \rangle \rightarrow 4$

Def A set of outcomes is called an event

$$E \subseteq S$$

$$Pr[E] = \sum_{s \in E} Pr[s]$$

Ex: When flipping 2-coins, what is the probability that at least one of them is heads.

$$S = \{H, T\}^2 = \{(H,H), (H,T), (T,H), (T,T)\}$$

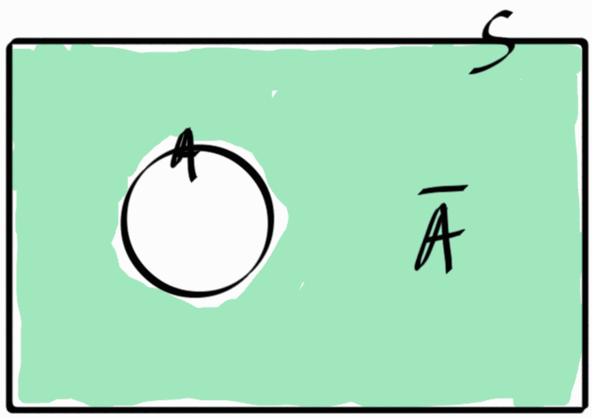
$$E = \{(H,H), (H,T), (T,H)\}$$

$$Pr[E] = Pr[(H,H)] + Pr[(H,T)] + Pr[(T,H)]$$

$$= 0.25 + 0.25 + 0.25 = 0.75$$

Theorem 10.4 (Properties of event probability)

Let S be your sample space and $A \subseteq S$, let $B \subseteq S$, and $\bar{A} = S - A$



$Pr[S] = 1$ $\quad$ $Pr[\emptyset] = 0$

$Pr[A] = 1 - Pr[\bar{A}]$

$Pr[\bar{A}] = 1 - Pr[A]$

Examples: when drawing 1 card from a 52-deck, What is the probability that the card we pick is <u>not</u> an Ace?

A - picking an Ace from the Deck

$\bar{A}$ - picking a card that is not an Ace

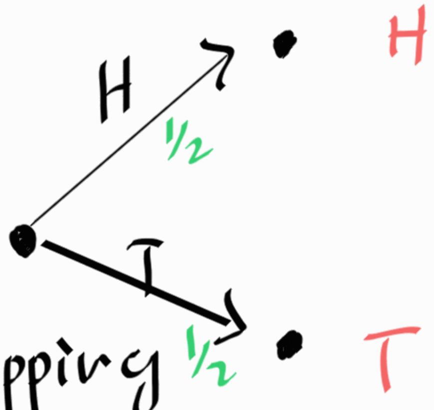
$Pr[A] = \frac{4}{52}$ $\qquad$ $Pr[\bar{A}] = \frac{48}{52}$

$$Pr[A] = 1 - \frac{48}{52} = \frac{4}{52}$$

Tree Diagrams in probability.

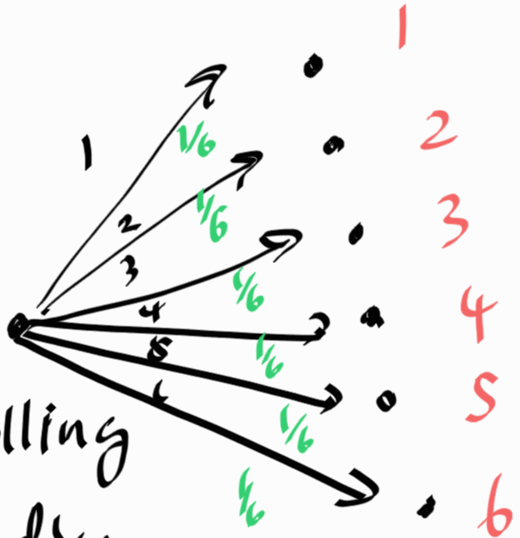
- Internal nodes of a tree diagram represents a random choice, labeled with the probability of each outcome
- leaves represents outcomes.

flipping a coin



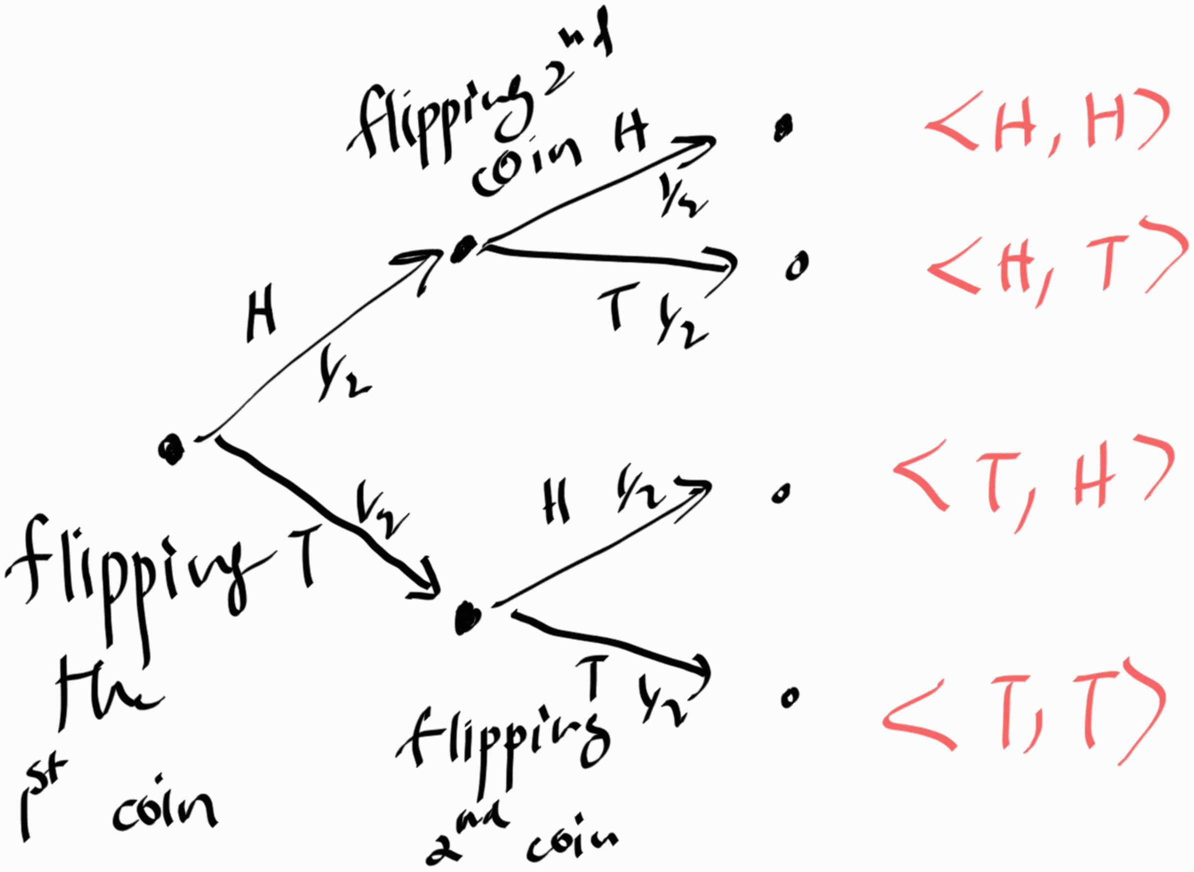
flipping a fair coin

rolling a dice



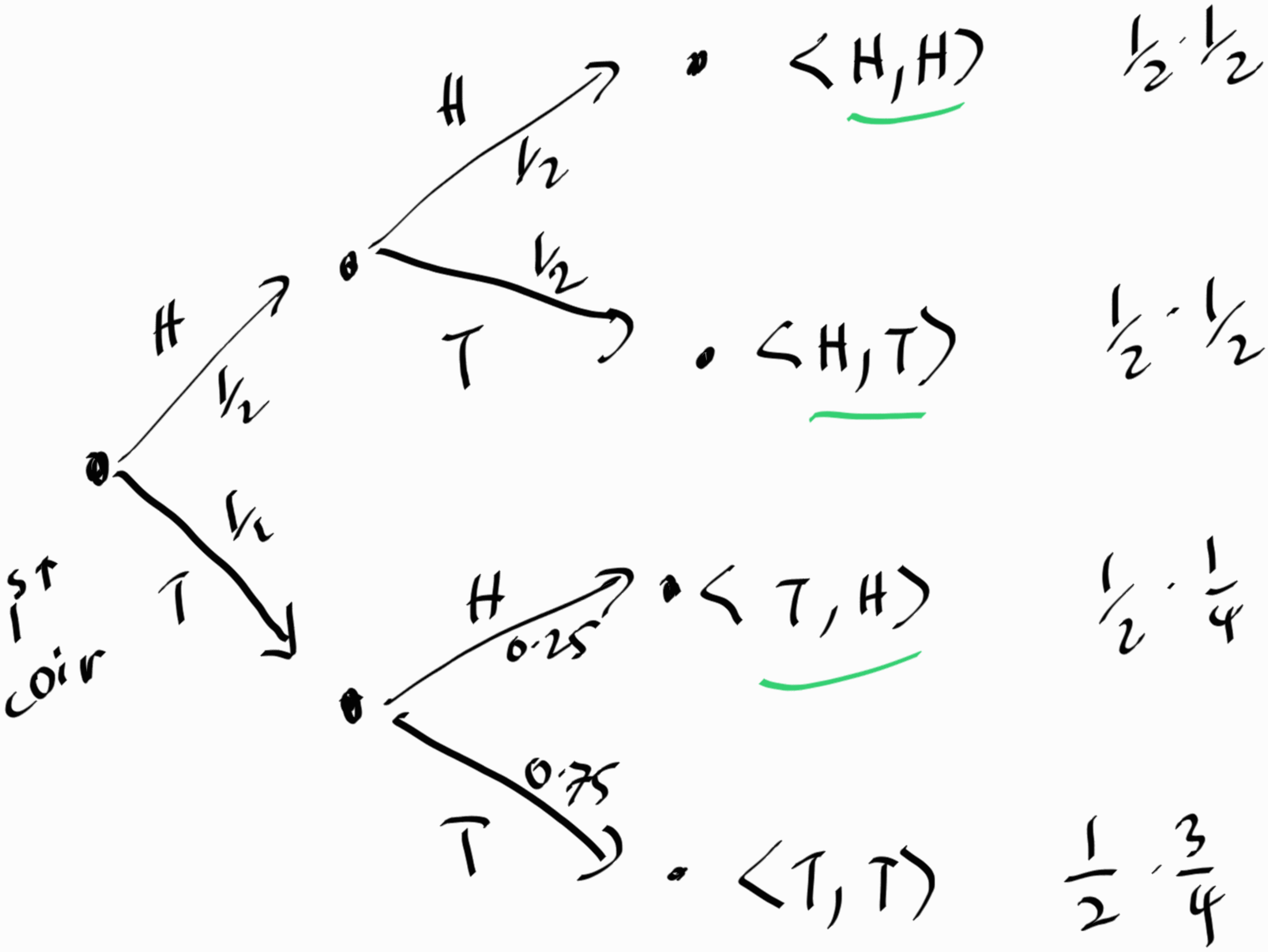
rolling a dice

flipping 2-coins



$$\Pr[\langle H,H\rangle] = \frac{1}{2} \cdot \frac{1}{2}$$

Example: flip 1st fair coin, If I get "H", then flip 2nd fair coin. If I get "T", flip an unfair coin with 0.75 probability of having T.



$$Pr[\langle T, T \rangle] = \frac{1}{2} \times \frac{3}{4} = \frac{3}{8}$$

$$\Pr[\text{at least one head}] = \frac{1}{2}\cdot\frac{1}{2} + \frac{1}{2}\cdot\frac{1}{2} + \frac{1}{2}\cdot\frac{1}{4}$$

$$= 5/8$$